

Seetharaman P

B.Tech CSE-AI Undergraduate

Chennai, India | +91 7010548108 | sri.p212004@gmail.com | [linkedin.com/in/seetharaman-p-b57176222](https://www.linkedin.com/in/seetharaman-p-b57176222) | github.com/sri21-cyber

TECHNICAL & SOFT SKILLS

Languages: Python, SQL, HTML, CSS, Matlab,
Libraries/Frameworks: TensorFlow, NumPy, Pandas, PyTorch,
Tools: Git, Microsoft Office, Video-Audio Editing
Soft Skills: Communication, Leadership, Active Listening, Meeting Deadlines

PROJECTS

- | | |
|---|---------------------|
| NLP-Based Fake News Detection | Apr 2023 – Jul 2023 |
| <ul style="list-style-type: none">– Developed an LSTM-based model to classify Tamil news articles, achieving 86.9% accuracy and reducing detection time by 20%.– Enhanced prediction accuracy by 10% through the use of TF-IDF, Bag of Words, and Word2Vec embeddings.– Performed hyperparameter tuning, improving overall model performance and reducing training time by 15%. | |
| Pneumonia Detection using UNet Model | Feb 2023 – May 2023 |
| <ul style="list-style-type: none">– Improved pneumonia detection accuracy by 30% using UNet, achieving a Dice coefficient of 0.95.– Reduced error rates by 15% and enhanced early diagnosis efficiency by 25%.– Applied advanced data augmentation, increasing generalization by 10% on unseen test data. | |
| Malayalam Hate Speech Detection | Jun 2023 – Aug 2023 |
| <ul style="list-style-type: none">– Achieved a 25% improvement in hate speech detection accuracy using speech-to-text classification with 92.51% accuracy.– Utilized MFCC, Mel Spectrogram, and Chromagram to extract robust features from audio data.– Reduced false positive rate by 20% through feature selection and model optimization techniques. | |
| Stress Detection using GRU and LSTM | Sep 2023 – Present |
| <ul style="list-style-type: none">– Developing a GRU and LSTM models, increasing stress prediction accuracy by 20% using HRV and ECG signals.– Optimized preprocessing techniques, lowering false negatives by 15% in stress prediction.– Optimized data preprocessing methodologies and extracted key features, leading to a significant improvement in model accuracy by 15%, ultimately increasing predictive reliability and providing valuable insights to stakeholders. | |

PUBLICATIONS

- | | |
|---|------------------------|
| Deep Learning Approach for Nepal Sign Detection | ICCIS Conference 2024 |
| <ul style="list-style-type: none">– Developed a high-accuracy Nepali Sign Language (NSL) gesture detection system using YOLOv8, achieving 98.6% accuracy on the NSL23 dataset.– Annotated a comprehensive NSL dataset, supporting scalability and real-time application of NSL recognition technology. | |
| Audio-based Hate Speech Detection in Malayalam using Machine Learning | ICDMAI Conference 2024 |
| <ul style="list-style-type: none">– Proposed a new dataset and method for hate speech detection in Malayalam, using audio features from social media.– Achieved 0.93 macro accuracy in binary classification and 0.80 in multi-class classification with a Random Forest Classifier. | |

EDUCATION

- | | |
|--|----------------------------------|
| Amrita Vishwa Vidyapeetham
<i>B.Tech in CSE-AI, CGPA: 7.3, Expected Graduation: May 2025</i> | Coimbatore, India
2021 – 2025 |
| NSN Memorial Senior Secondary School
<i>AISSCE, CGPA: 9.47</i> | Chennai, India
2019 – 2021 |

CERTIFICATIONS

Deep Learning Specialization by DeepLearning.AI
Machine Learning by Stanford University (Coursera)
AWS Cloud Practitioner Essentials
NLP with Python by Udemy